

HAIFA REFINERIES, Israel

WMO: 401550

Lat: 32.8034N

Lon: 35.0548E

Elev: 28

StdP: 14.68

Time Zone: 2.00 (ISR)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	45.4	47.5	20.4	15.3	66.9	26.6	20.6	63.8	24.8	62.0	22.0	60.1	5.3	150	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.7	91.5	70.6	89.2	73.5	87.8	73.7	78.2	85.6	77.3	85.2	76.5	84.5	10.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.8	135.2	83.4	74.8	130.4	82.9	73.8	126.1	82.7	41.7	85.9	40.8	85.5	40.1	84.6	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.5	17.7	15.5	DB	41.0	101.1	2.6	3.8	39.1	103.8	37.6	106.0	36.1	108.1	34.2	110.8	
			WB	37.9	80.0	2.9	1.8	35.8	81.3	34.1	82.3	32.4	83.3	30.3	84.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.5	56.9	57.3	60.3	65.8	71.4	77.2	81.6	82.2	79.5	74.2	67.0	59.5	(6)
	DBStd	10.03	3.87	4.18	4.91	6.08	4.73	2.94	1.97	2.04	2.59	4.49	5.30	3.99	(7)
	HDD50	5	2	2	0	0	0	0	0	0	0	0	0	1	(8)
	HDD65	909	251	220	166	52	4	0	0	0	0	2	38	176	(9)
	CDD50	7113	217	205	319	474	664	816	979	999	884	750	511	295	(10)
	CDD65	2543	2	3	20	77	203	366	514	534	434	286	98	6	(11)
	CDH74	23587	1	11	124	568	973	2901	5773	6365	4408	2063	393	7	(12)
	CDH80	7001	0	2	31	218	250	629	1971	2321	1129	397	53	0	(13)
(14) Wind	WSAvg	7.1	7.7	7.7	7.5	7.7	7.1	6.8	6.5	6.5	6.8	6.4	7.2	7.8	(14)
(15) Precipitation	PrecAvg	24.2	6.2	4.5	2.2	0.9	0.2	0.0	0.0	0.0	0.1	1.2	3.4	5.2	(15)
	PrecMax	34.6	12.6	10.9	6.5	3.5	1.0	0.3	0.5	0.1	0.6	3.6	10.7	11.5	(16)
	PrecMin	11.7	1.7	1.1	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.8	(17)
	PrecStd	6.5	2.7	2.4	1.6	0.9	0.2	0.0	0.1	0.0	0.2	1.1	2.6	2.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.5	74.7	84.2	93.5	95.0	91.2	93.2	91.9	91.7	94.5	85.1	74.2	(19)
		MCWB	56.4	55.3	58.9	62.5	63.4	71.0	74.2	77.4	68.4	63.5	61.5	56.8	(20)
	2%	DB	68.1	69.8	76.8	86.6	85.9	87.0	90.1	89.6	87.9	87.6	80.9	70.9	(21)
		MCWB	54.3	54.1	57.6	61.5	63.6	71.7	74.7	76.5	74.3	65.2	60.7	56.9	(22)
	5%	DB	65.9	66.6	72.3	80.7	81.6	85.0	88.6	88.5	86.2	83.4	78.1	68.8	(23)
		MCWB	55.0	54.2	57.2	60.4	65.1	71.9	74.7	75.9	73.9	69.6	61.1	56.0	(24)
	10%	DB	64.0	64.4	68.0	75.1	78.6	83.4	87.0	87.3	84.8	81.5	75.8	66.7	(25)
		MCWB	54.5	54.8	57.9	59.7	66.5	71.4	74.5	75.3	73.3	70.4	61.8	55.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.1	60.3	64.0	67.3	71.6	75.9	79.1	80.1	78.4	76.7	70.0	62.5	(27)
		MCDB	66.6	66.8	72.9	80.9	78.3	82.4	86.4	87.4	85.0	82.2	76.9	68.7	(28)
	2%	WB	58.3	58.5	62.1	64.7	70.3	74.7	77.5	78.4	77.1	74.6	67.7	60.2	(29)
		MCDB	64.2	65.5	69.6	74.2	77.2	82.2	85.4	85.9	84.1	81.0	74.7	66.7	(30)
	5%	WB	56.9	57.2	60.7	63.6	69.3	73.8	76.5	77.7	76.0	73.4	66.0	58.5	(31)
		MCDB	63.3	63.7	67.3	72.5	76.4	81.5	84.8	85.6	83.5	80.2	73.9	65.2	(32)
	10%	WB	55.5	55.7	59.3	62.5	68.3	72.9	75.9	77.0	75.0	71.9	64.3	57.3	(33)
		MCDB	62.1	62.5	66.2	71.3	75.6	80.8	84.2	85.3	82.9	79.1	72.4	64.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	12.9	13.3	13.8	13.1	11.9	11.5	10.7	11.0	13.3	14.6	12.3	(35)
		MCDBR	14.7	16.5	20.4	24.5	19.7	14.6	13.9	11.9	12.7	16.6	17.5	15.0	(36)
	5% WB	MCWBR	8.1	9.4	9.3	8.8	8.1	6.0	5.3	4.4	6.0	9.1	9.2	7.6	(37)
		MCWBR	12.8	14.1	14.6	15.9	14.4	13.3	12.4	11.0	12.3	12.8	14.6	12.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.387	0.419	0.459	0.503	0.499	0.432	0.450	0.455	0.438	0.449	0.405	0.389	(40)
		taud	2.273	2.164	2.050	1.948	2.000	2.237	2.197	2.183	2.232	2.172	2.265	2.280	(41)
	Ebn at Noon		258	260	259	253	255	271	266	263	262	247	249	249	(42)
		Edh at Noon	35	43	51	59	56	44	46	46	42	42	35	33	(43)
(44) All-Sky Solar Radiation	RadAvg		891	1157	1615	1993	2341	2572	2498	2273	1937	1453	1067	830	(44)
	RadStd		64	102	98	109	98	44	28	41	49	68	56	64	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (17 neighbors)		+0.79	N/A	-3.46	N/A	N/A	N/A	N/A	-126	+288	+193	

Nomenclature: See separate page